

Chung-Yen (Matt) Chiang

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Location: Taipei, Taiwan

Professional Experience

Amazon - Connectivity Team

Apr 2026 – Present

Software Development Engineer II

- Led wireless bring-up (Wi-Fi/BLE) for embedded systems on **FreeRTOS**, consolidating chipset BSPs/SDKs across next-gen ASIC and wireless vendors for NPI projects.

Amazon - Connectivity Team

Aug 2023 – Mar 2026

Software Development Engineer I (L4)

- Developed a **modular BLE setup subsystem** from scratch, incorporating an **E2E encryption channel** and dynamic country code configuration for global SKU deployment.
- Optimized DHCP reconnection logic, achieving a **76% reduction in latency** and 69% performance gain in unstable network environments.
- Fine-tuned **WMM QoS (EDCA)** parameters to prioritize latency-sensitive traffic for high-fidelity streaming services.
- Implemented advanced Wi-Fi power-saving mechanisms and wake-up patterns, significantly extending battery-operated IoT device longevity.

Amazon - Connectivity Team

Sep 2022 – Jul 2023

Software Development Engineer Intern

- Refined priority-based connection management logic to improve connectivity consistency across diverse IoT ecosystems.
- Debugged and resolved 20+ critical firmware issues during the validation phase of major product launches.
- Developed automated diagnostic tools for interoperability testing with 60+ commercial router models.

Academia Sinica – Institute of Information Science

Jul 2022 – Aug 2022

Research Intern (Advised by Prof. Meng Chang Chen)

- Reverse-engineered obfuscated malware using **IDA Pro** to identify sandbox evasion triggers and anti-analysis behaviors.
- Researched hardware-level evasion techniques (CPU/Registry checks) to improve the detection accuracy of analysis environments.
- Implemented a modular detection framework in C and Python to enhance the flexibility of the **Cuckoo Sandbox** monitor.

Research & Projects

Implementation Analysis of Secure Channels for Smart Manufacturing Systems in Cloud Environments (Advised by Prof. Nai Wei Lo)

Jul 2021 – Feb 2022

1st Author - Cryptology and Information Security Conference 2022

1st Author - Ministry of Science and Technology (MOST) Scholarship

- Awarded the **Undergraduate Research Scholarship** by the Ministry of Science and Technology for excellence in secure systems research.

- Researched and implemented **Site-to-Site IPsec** secure channels for industrial communication between local gateways and AWS VPC.
- Conducted deep packet inspection (DPI) with **Wireshark** and benchmarked performance using iPerf3 and Netperf.
- Investigated **ARMv8 Cryptographic Extensions** to accelerate encryption/decryption on resource-constrained devices.

Billiards Guidance with HoloLens 2 (Advised by Prof. Chuan Kai Yang) *Jul 2021 – May 2022*

Team Developer — Special Topics in Information Management

- Built a real-time trajectory projection system on HoloLens 2 using BFS-based reflection algorithms to simulate and visualize cue ball paths in MR.

Technical Skills

Core Domains	Embedded Systems, Wireless Protocols (Wi-Fi/BLE), RTOS (FreeRTOS), Deep Packet Inspection
Programming	C, C++, Python, Shell Scripting
Tools	Linux, Wireshark, Git, JTAG/GDB
Language	English (IELTS 6.5, TOEIC 800), Mandarin (Native)

Education

National Taiwan University of Science and Technology *2019 – 2023*

B.B.A. in Information Management

GPA: 4.01/4.30

Academic Excellence Award (Ranked **Top 2** of class for three consecutive semesters)

Selected Coursework: Computer Architecture (A+), Data Structures (A+), Operating Systems (A), Algorithms (A)